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PYTHON PROGRAMMING

* Introducing Python -

→ Python is a high-level, general-purpose, and very popular programming language.
→ It is being used in web development, Machine learning applications, etc.
→ Python programs are generally smaller than the other programming languages.
→ Programmers have to type relatively less and the indentation requirement of the language, makes them readable at the time.

→ Very Simple Syntax. & Easy to learn.

```
print("Hello")
```

→ General purpose language (Simple yet very powerful language)
- → Console Application & Scripts
- → Desktop Application
- → Web Application
- → Game Development
- → Machine Learning, Deep learning, AI, Big data etc

→ Multiparadigm Support
- → Procedural Style Programming like C.
- → OOPs like Java.
- → Functional programming like list.

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→ Portable or Platform Independent

- Programs are typically first compiled into an intermediate code, then the code is run by the interpreter.

→ Dynamically Typed

```
x = 10
y = "geeks"
x = "python"
```

This feature is different from C++ & Java. They are statically Typed.

In python we don't have to specify the variable name.

But the disadvantage is in this the type is assigned in the run time of the program, so there are more chances of run time error.

→ Automatic Garbage Collection

It means when you dynamically allocate memory so you don't have to worry to realising this memory.

→ Popular Application Built in Python

Youtube, Netflix, Quora, Instagram, DropBox

⇒ print () in Python –

A function is a set of instructions that take some input from us, do some work on those

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parameters and produce some output.

Example - print ("Hello")
print ("Welcome", "to", "GfG")
print ()
print ("Hope you are enjoying Python")

Output - Hello
Welcome to GfG

Hope you are enjoying Python

→ ~~end~~ end and sep parameters in print()

Eg - print ("Welcome", end = " ")
print (" to GfG")

Output - Welcome to GfG

End parameter creates space and it tells what should be printing after its printing.
And in sep parameter when we have multiple input parameters we use sep to separate them.
By default they are separated by space.

Eg - print ("10", "04", "2023", sep = "-")

Output - 10-04-2023

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⇒ Variables in Python -

Variable is a name that is used to refer to memory location. Python variable is also known as an identi-fier and used to hold value.

In python, we don't need to specify the type of variable because Python is a infer language and smart enough to get variable type.

Example -

```
price = 100
tax = 18
total-price = price + tax
print (total-price)
```

Output - 118

How variables work ?

```
x = 10
y = "geeks"
z = 20
```

[10] ← x
[geeks] ← y
[20] ← z

if we say, w = z
then [20] ← z, w

but if, w = 30 [30] ← w

```
print (x)
print (y)
print (z)
print (w)
```

O/P -

```
10
geeks
20
30
```

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Python is Dynamically Typed

```
x = 10
print(x)
x = "geeks"
print(x)
```

O/P - 10
geeks

NOTE - Using a variable without assigning it cause error.

Eg - print(x)
O/P - NameError : name 'x' is not assigned

⇒ input() in Python -

This function is used to take input from the user

Example -
```
name = input ("Enter the name =")
print (" welcome" + name)
```

O/P - Enter the name = Himani
Welcome Himani

Example - # Python program for addition

```
x = input (" Enter First Number :")
y = input (" Enter Second Number :")
x = int (x)
y = int (y)
res = x + y
print (" Sum is", res)
```

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Output - Enter First Number = 10
Enter Second Number = 20
Sum is 30

NOTE - input() always gives us a string so whenever we need mathematical expressions we have to convert it from string.

=> Type() in Python -

Type is a built-in function that tells you data type of a variable or a value.

Eg -
```
a = 10
print (type (a))
b = 10.5
print (type(b))
c = 2 + 3j
print (type (c))
```

O/P -
```
< class, 'int' >
< class, 'float' >
< class, 'complex' >
```

- In python we use "None" to indicate that we don't know the value yet or we didn't assign any value to this variable.

In python there is no char type, if you want char you should create single char string. (String is a sequence of char)

Eg -
```
str = "gfg"
print (type (str))
J = [10, 20, 30]
print (type (J))
```

(array and dynamically add and remove items. Items are stored in continuous locations)

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